Embeddings

Embeddings

Example Studies

Example 1: Pseudoword meanings

Please provide a short definition for the following word:

kittings

tarvos

sibre

teziels

de Varda, Gatti, Marelli, & Günther (2024, *Computational Linguistics*)

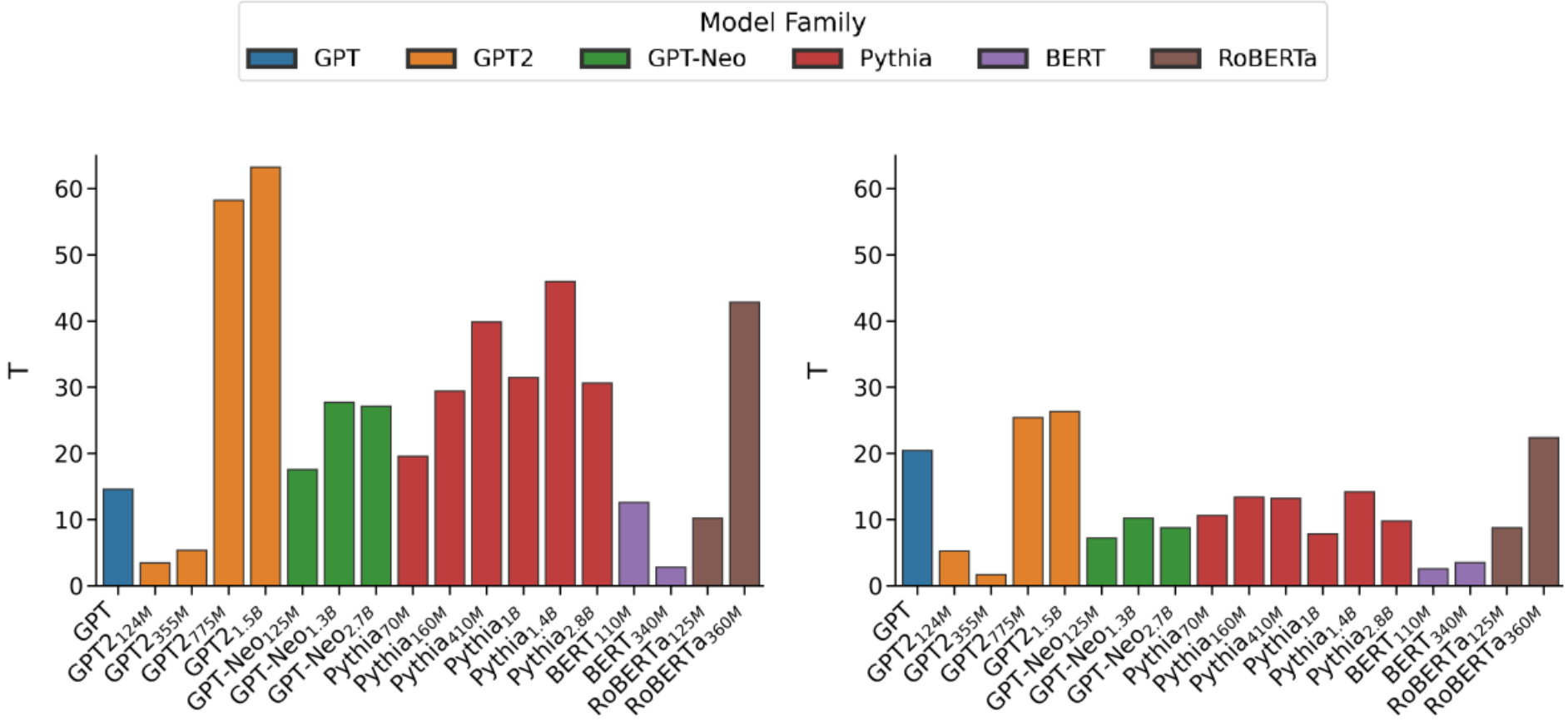
Example 1: Pseudoword meanings

Please provide a short definition for the following word:

word	definition	cosine
kittings	Strings or ropes used to hold a balloon down	.399
	Small feline animals born in a barn	.497
tarvos	Flooring made of wooden planks	.302
	A region of greece	.533
sibre	[sibre] is a form of animal with a lot of fur	.389
	A flamboyant hat	.593
teziels	A food that is shaped like a pretzel but is something else	.391
	A pebble with irregular sides	.548

de Varda, Gatti, Marelli, & Günther (2024, *Computational Linguistics*)

Example 1: Pseudoword meanings



(a) Word data (b) Pseudoword data

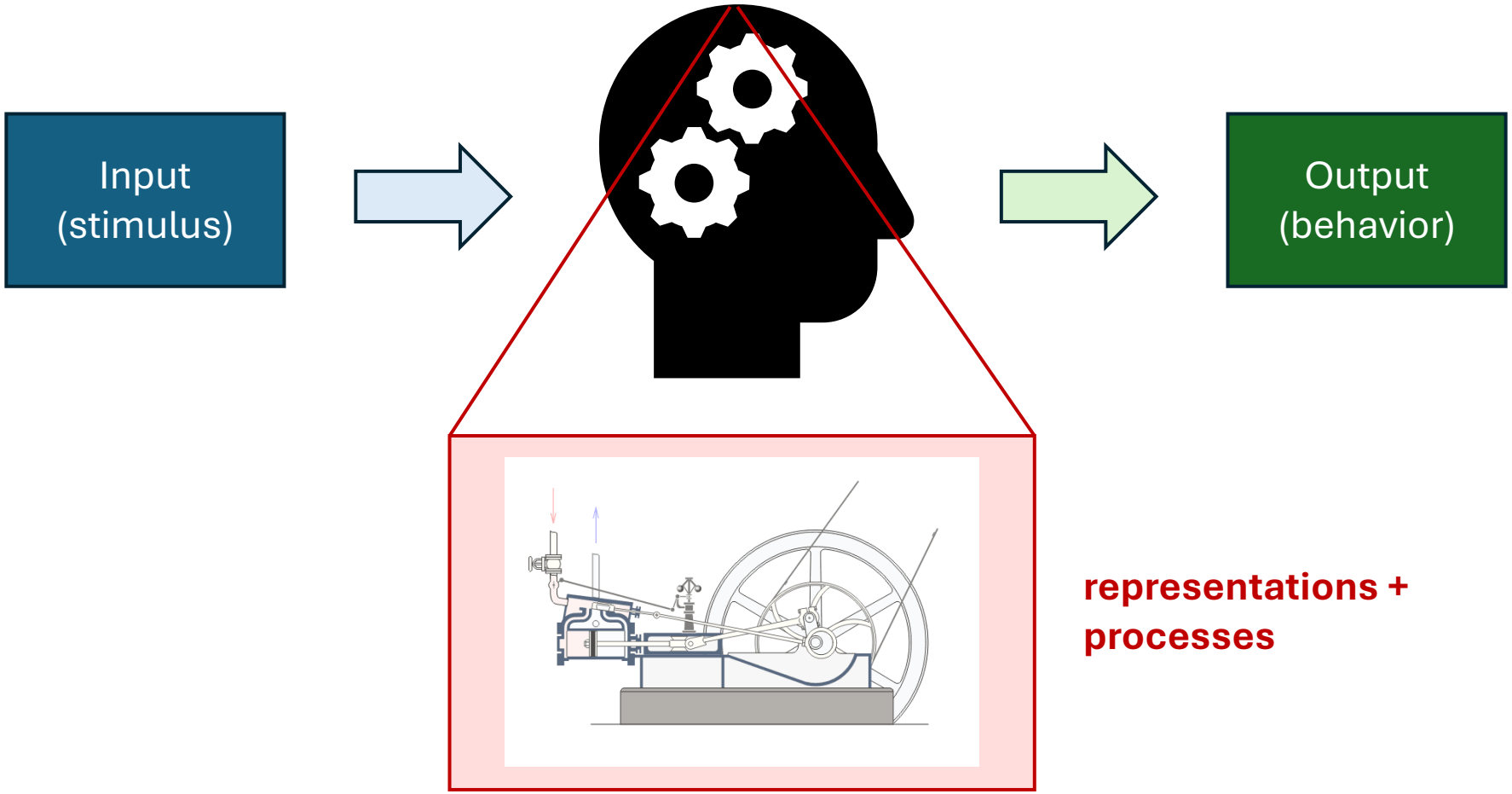
T: Difference in embedding similarity (LMM coefficient) between

- **matching** definitions (produced for the **respective** target word)
- **mismatching** definitions (produced for **other** target words)

de Varda, Gatti, Marelli, & Günther (2024, *Computational Linguistics*)

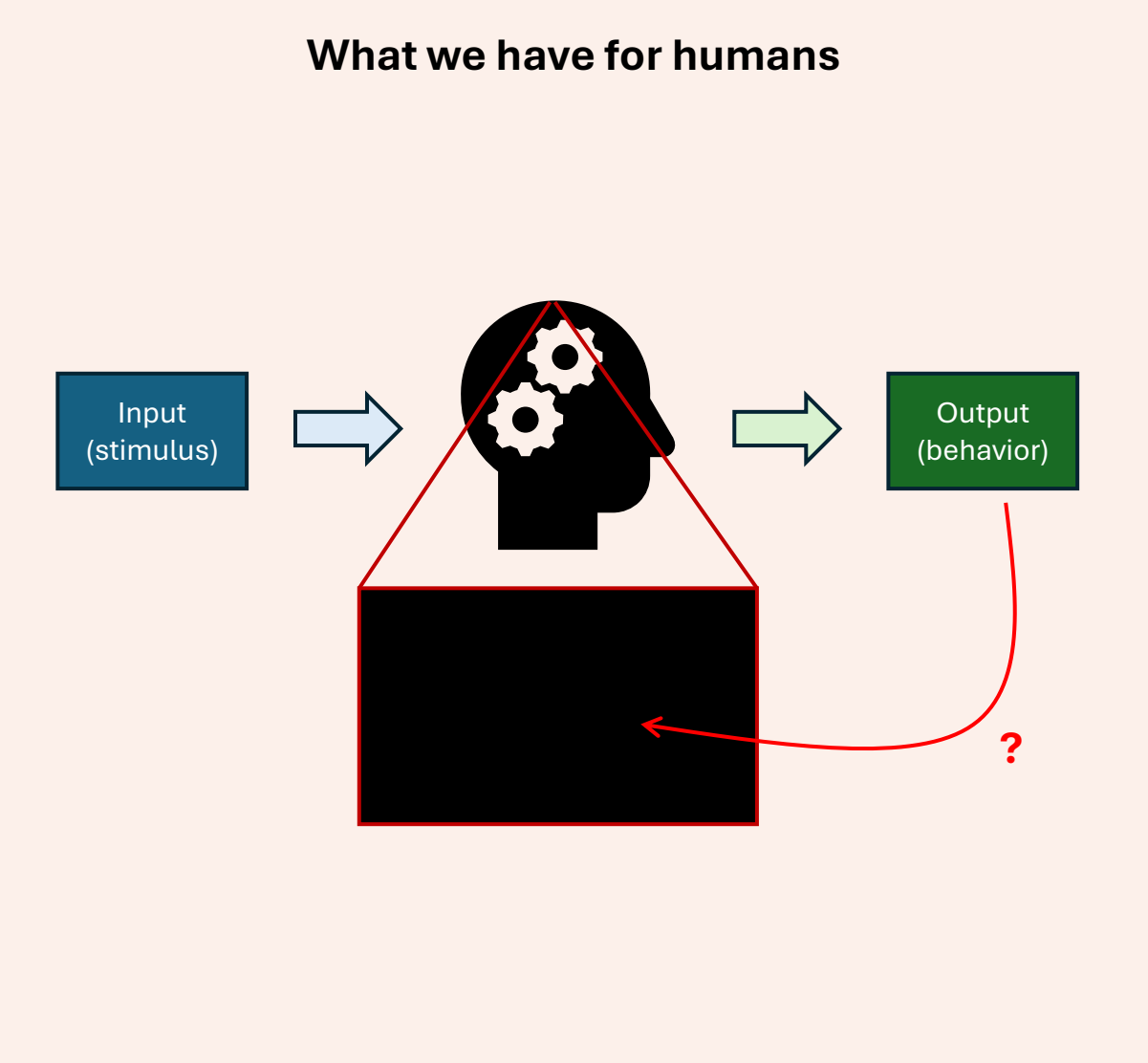
Example 2: Recovering LLM embeddings from their behavior

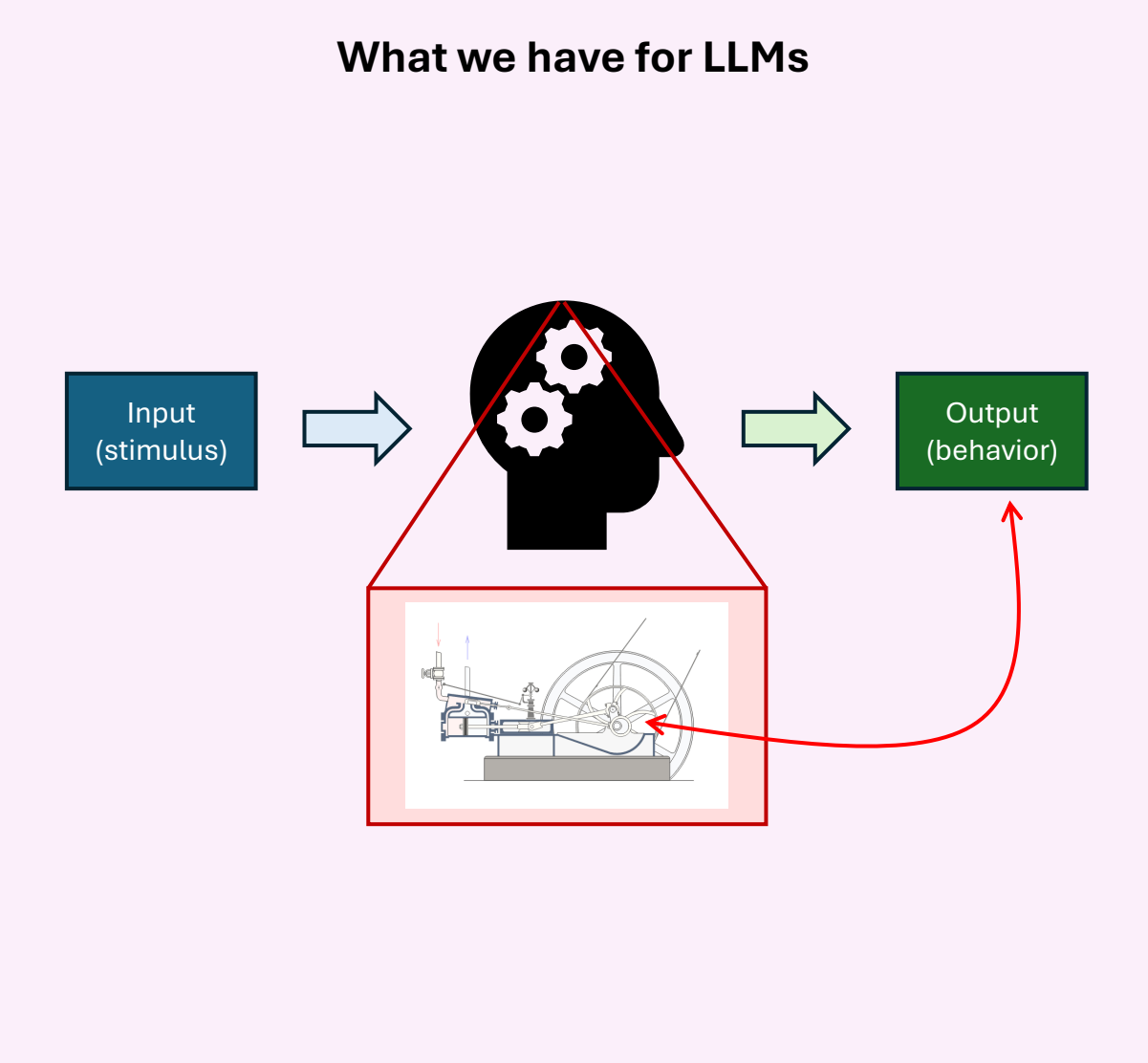
What we try to do in cognitive psychology



Schiekiera, Zimmer, Roux, Pokutta, & Günther (2026, *arXiv*)

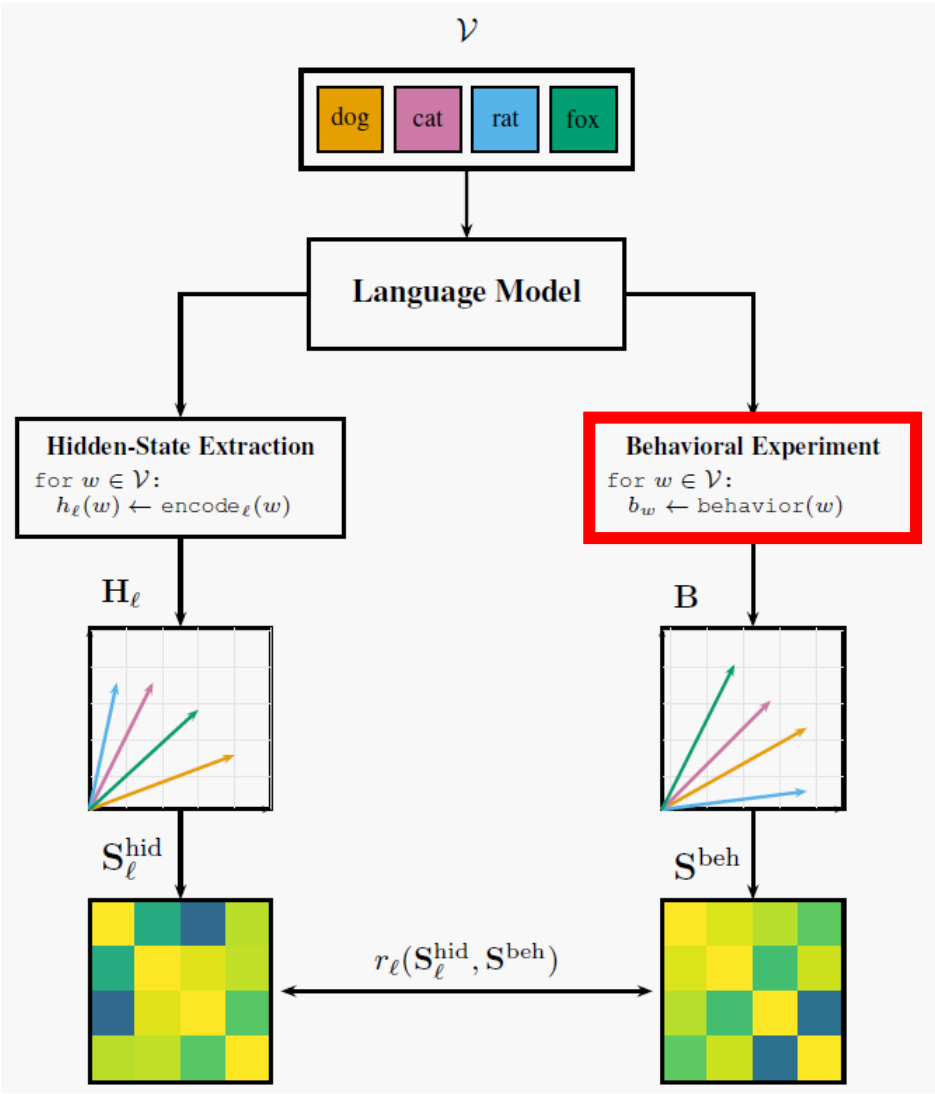
Example 2: Recovering LLM embeddings from their behavior

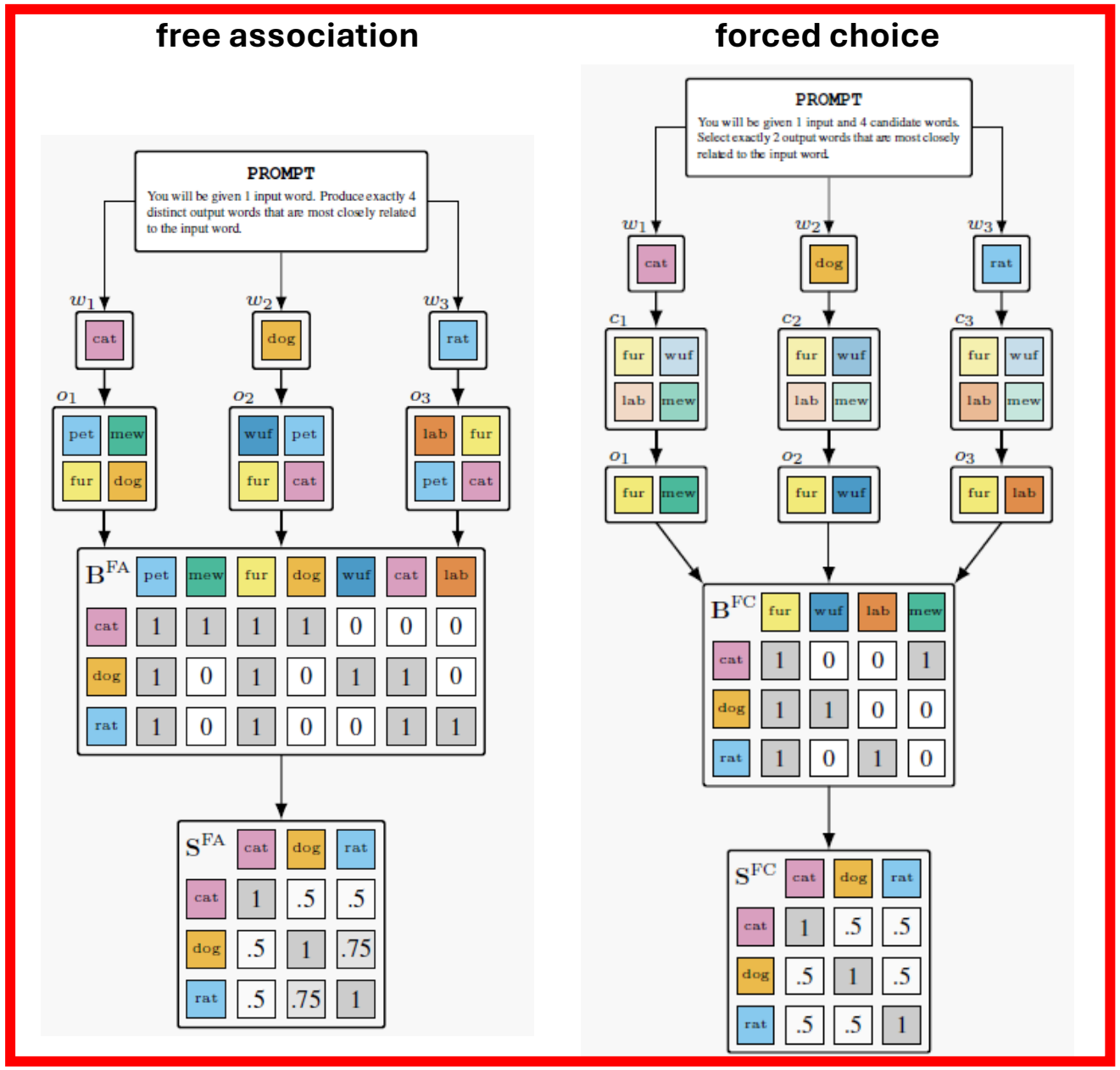




Schiekiera, Zimmer, Roux, Pokutta, & Günther (2026, *arXiv*)

Example 2: Recovering LLM embeddings from their behavior





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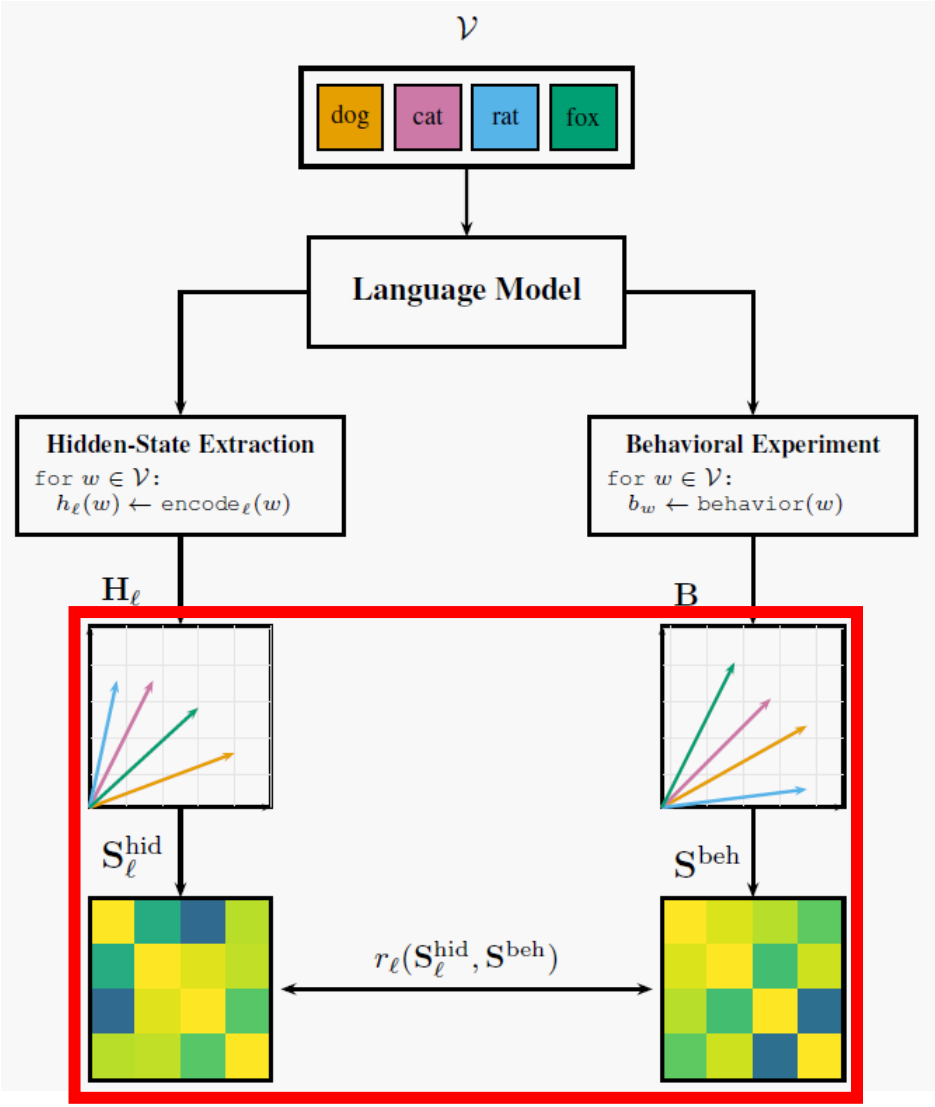
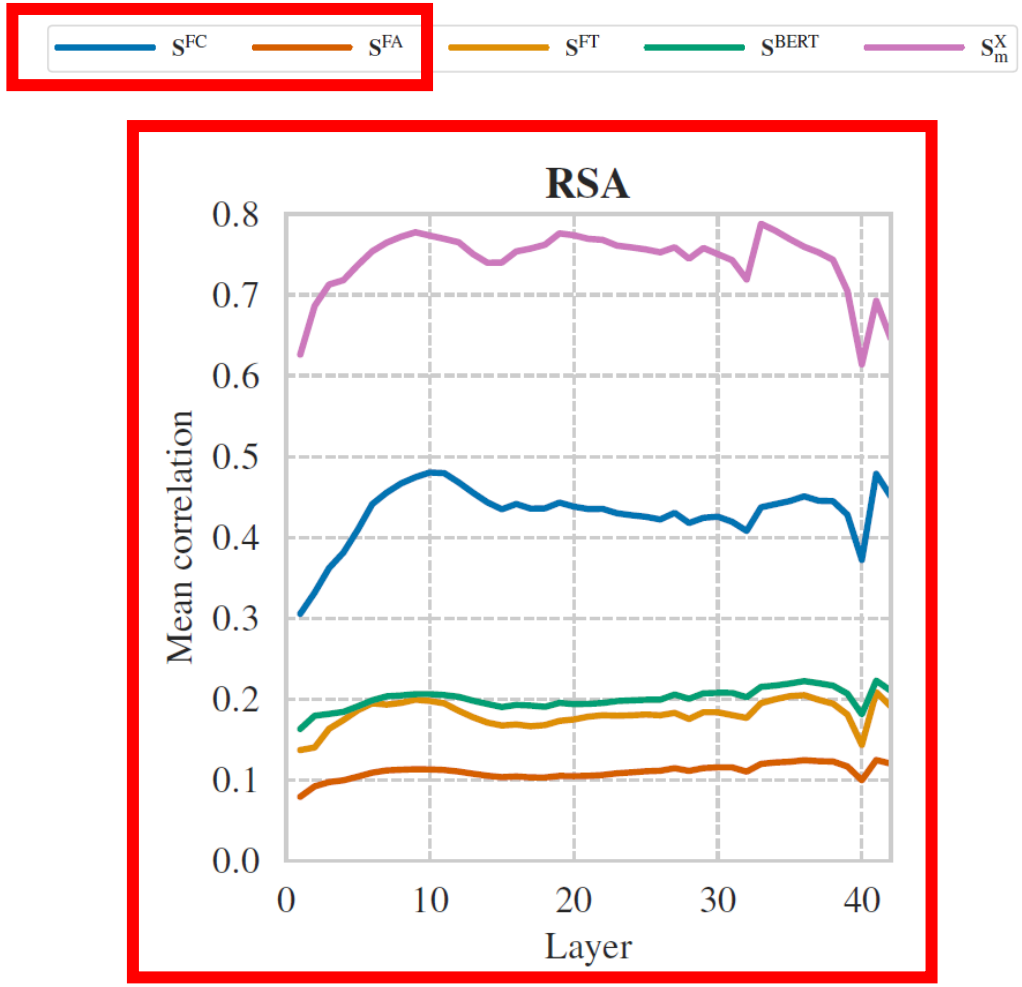
Example 2: Recovering LLM embeddings from their behavior





Schiekiera, Zimmer, Roux, Pokutta, & Günther (2026, *arXiv*)

Embeddings

Extracting embeddings in R

Extracting embeddings in R

- Extracting embeddings is often referred to as *feature extraction*

- R is *not* the native language for extracting embeddings (and LLM interactions in general) – it‘s all designed for python

  - For python instructions: start from https://huggingface.co/tasks/feature-extraction
    - HuggingFace is *the* main platform for LLMs
  - For a tutorial, see Hussain et al. (2024, *Behavior Research Methods*)
  - If you want to get really serious about working with LLMs, this is where you eventually want to go

Extracting embeddings with reticulate

- Here, we are using the R package **reticulate** to integrate python in R
  - This package provides an **interface** to the relevant python libraries
  - This can make installing the package a bit of a hassle (somewhat depending on your OS)

Extracting embeddings: The text package

- On the subsequent slides, you will see some examples for the R package **text**
- Interaction is more intuitive and output is nicer than for **reticulate**
- Installing is more of a hassle (but only has to be done once!)
- For detailed instructions, see https://r-text.org/articles/Extended_Installation_Guide.html

Kjell, Giorgi, & Schwartz (2023, *Psychological Methods*)

Extracting embeddings: The text package

https://r-text.org/articles/Extended_Installation_Guide.html

This is where you have to install a lot of things (incl. Visual Studio for C++ on Windows...)

```
### install the text package (once!)
### requires an up-to-date version of R (> 4.4.3)
# install.packages("text")

## for more info: https://r-text.org/articles/Extended_Installation_Guide.html

### load the text package (every session)
library(text)
library(reticulate)

##### conda environment

### Install text required python packages in a conda environment (with defaults)
text::textrpp_install()

### Show available conda environments.
reticulate::conda_list()

### Initialize the installed conda environment.
###    save_profile = TRUE saves the settings so that you don't have to run
###    textrpp_initialize() after restarting R.
text::textrpp_initialize(save_profile = TRUE)
```

Kjell, Giorgi, & Schwartz (2023, *Psychological Methods*)

Extracting embeddings: basics

```r
# Example text
texts <- c("I'm feeling relatedness with
 others", "That's great!")

# Defaults
embeddings <- textEmbed(texts)
```

Embeddings for all individual tokens in the text

```r
# Output
embeddings$tokens

$texts
$texts[[1]]
A tibble: 10 x 771

tokens          Dim1          Dim2          Dim3
<chr>           <dbl>         <dbl>         <dbl>
[CLS]           0.467         0.0459        -0.222
i               0.593         0.198         -0.384
`               0.0611        0.0160        -0.0248
m               0.236         0.231         -0.552
feeling        -0.402         0.0777         0.252
related         0.230        -0.400          0.479
#ness          -0.695        -0.328          0.782
with            0.264         0.219          1.11
others         -0.298         0.0784         0.139
[SEP]           0.0446        0.0103        -0.0345

# ... with 765 more variables: Dim4 <dbl>, Dim5 <dbl>,
 Dim6 <dbl>, ...

 ...
```

Default: second-to-last layer (here: layer 11)

Usage

```r
textEmbed(
  texts,
  text_ids = NULL,
  group_ids = NULL,
  model = "bert-base-uncased",
  layers = -2,
  dim_name = TRUE,
  aggregation_from_layers_to_tokens = "concatenate",
  aggregation_from_tokens_to_texts = "mean",
  aggregation_from_tokens_to_word_types = NULL,
```

Embedding for the entire text

```r
# Output
embeddings$texts

$texts
$texts[[1]]

A tibble: 10 x 771

Dim1_texts <dbl>     Dim2_texts <dbl>     Dim3_texts <dbl>
0.0501                0.0149               0.155
0.0942               -0.104               -0.0625
# ... with 765 more variables: Dim4_texts <dbl>,
   Dim5_texts <dbl>, Dim6_text <dbl>, ...

...
```

Kjell, Giorgi, & Schwartz (2023, *Psychological Methods*)

Extracting embeddings: different layers

```
# Get hidden states for "I am fine"
imf_embeddings_11_12 <- textEmbedRawLayers(
  "I am fine",
  layers = 11:12
)
imf_embeddings_11_12

# OUTPUT


$context_tokens
$context_tokens$texts
$context_tokens$texts[[1]]
# A tibble: 10 x 771
tokens   token_id   layer_number   Dim1      Dim2       Dim3
 <chr>     <int>      <int>       <dbl>     <dbl>      <dbl>
[CLS]        1         11          0.266   -0.0910   -0.0521
i            2         11          0.178   -0.296     0.188
am           3         11         -0.114   -0.542     0.0757
fine         4         11          1.18    -0.434    -0.317
[SEP]        5         11          0.0461   0.0134   -0.0311
[CLS]        1         12          0.0609   0.377     0.213
i            2         12         -0.173    0.0227    0.0647
am           3         12         -0.406   -0.319     0.128
fine         4         12          0.763   -0.373    -0.009
[SEP]        5         12          0.798   -0.0405   -0.324
# ... with 765 more variables: Dim4 <dbl>, Dim5 <dbl>, Dim6
 <dbl>, ...
```

Kjell, Giorgi, & Schwartz (2023, *Psychological Methods*)

Extracting embeddings: different layers

```
# 1. Concatenate layers (results in 1,536 dimensions).
textEmbedLayerAggregation(
  imf_embeddings_11_12$context_tokens,
  layers = 11:12,
  aggregation_from_layers_to_tokens = "concatenate",
  aggregation_from_tokens_to_texts = "mean"
)
# OUTPUT

$x
# A tibble: 1 × 1,536
 Dim1      Dim2       Dim3       Dim4      Dim5      Dim6
<dbl>     <dbl>      <dbl>      <dbl>     <dbl>     <dbl>
0.311    -0.270    -0.0274    -0.408     0.202     0.113
...with 1,530 more variables: Dim7, Dim8, Dim9, ...

# 2. Only select layer 11 (768 dimensions).
textEmbedLayerAggregation(
  imf_embeddings_11_12$context_tokens,
  layers = 11,
  aggregation_from_tokens_to_texts = "mean"
)

# OUTPUT

$x
# A tibble: 1 × 768
 Dim1      Dim2       Dim3       Dim4      Dim5      Dim6
<dbl>     <dbl>      <dbl>      <dbl>     <dbl>     <dbl>
 0.311    -0.270    -0.0274    -0.408     0.202     0.113
...with 762 more variables: Dim7, Dim8, Dim9, ...
```

Kjell, Giorgi, & Schwartz (2023, *Psychological Methods*)